Unsupervised Dialect-Aware Phonetic Contrastive Learning for Bengali-English Code-Mixed Text

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Abstract

Code-mixed text, especially Bengali-English content commonly found on social media, presents significant challenges for traditional NLP systems. The lack of standardized spelling, frequent phonetic variations influenced by regional dialects, and the scarcity of annotated datasets make supervised approaches difficult to scale effectively.In this work, we propose an unsupervised dialect-aware phonetic contrastive learning framework designed to learn meaningful word representations directly from raw code-mixed data. The core idea is to incorporate phonetic information by mapping different transliterated forms of the same word to a common phonetic representation, and then using contrastive learning to bring similar-sounding words closer while separating unrelated ones in the embedding space.
Our experiments, carried out as part of the SLP Phase 5 implementation, show that the model successfully captures dialectal variations and reduces the effect of inconsistent spellings. It also demonstrates strong generalization to previously unseen word forms. On a dataset of 46,369 sentences, the model achieves a silhouette score of 0.5161, a Davies-Bouldin index of 0.6989, and a Calinski-Harabasz score of 55,320.7, along with an Adjusted Rand Index of 0.8089 when compared with the original dataset labels.Overall, the results suggest that phonetic-aware contrastive learning is an effective and scalable solution for handling low-resource, dialect-rich code-mixed languages, with potential applications in tasks such as sentiment analysis, language identification, and hate speech detection.

1. Introduction

The rapid growth of user-generated content on platforms such as Facebook, YouTube, and Twitter has made code-mixed language a common feature of digital communication in South Asia. Bengali speakers frequently combine Bengali and English within the same sentence, often writing everything in the Roman script — a form popularly referred to as *Romanised Banglish*. This type of text is highly variable, not only because it lacks a standard writing system, but also because it reflects the influence of multiple regional dialects such as Dhaka, Chittagonian, Sylheti, Barishal, Noakhali, and Rajshahi. As a result, the same word can appear in several different spellings depending on pronunciation and regional usage.

Most existing NLP systems are designed for clean, monolingual text, and they struggle significantly when applied to code-mixed data. There are a few key reasons for this. First, a single word may appear in many different forms that

sound the same but are spelled differently. Second, standard tokenization techniques are not well-suited for handling language switching within or across words. Third, multilingual models often underrepresent Bengali, leading to what is commonly referred to as the *embedding barrier*, where the language does not receive sufficient representation in the model's vocabulary. While supervised fine-tuning can improve performance to some extent, it relies heavily on large, well-annotated datasets, which are extremely limited for code-mixed and dialectal Bengali text.

To address these challenges, we take a fully unsupervised approach. The key contributions of this work are as follows:

- **Phonetic Encoding Module:** We introduce a phonetic encoding mechanism that converts Romanised Bengali words into a unified phonetic form. This helps reduce spelling variation while preserving the underlying pronunciation.
- **Dialect-Aware Contrastive Learning:** We design a contrastive learning objective that uses phonetic similarity to construct positive and negative pairs. This allows the model to learn embeddings that are robust to dialectal and spelling differences without requiring labeled data.
- **End-to-End Unsupervised Framework:** We develop a complete pipeline and evaluate it on a dataset of 46,369 Bengali-English code-mixed sentences. The model achieves strong clustering performance, including a silhouette score of 0.5161 and an Adjusted Rand Index of 0.8089.
- **Interpretable Dialect Clusters:** The learned representations naturally form meaningful clusters corresponding to Pure Bengali, Code-Mixed, and Formal Bengali text. These clusters show clear statistical separation, as indicated by a high CMI-based F-score (14,992.81), highlighting the effectiveness of the approach.

2. Related Work

2.1 Code-Mixed NLP and Dialect Modelling

As code-mixed data has become more common, research in this area has grown significantly. Joshi et al. (2025) provide a detailed survey of dialect-aware NLP methods and point out an important issue: most existing benchmarks do not adequately represent non-standard language varieties. This leads to biased evaluations and limits how well models perform in real-world settings.

Building on this, Sheth et al. (2026) examine the performance of modern large language models on code-switched data. Their findings show that even advanced models struggle in zero-shot settings and that evaluation benchmarks are often inconsistent across different language pairs. Together, these studies highlight the need for models that can naturally handle dialectal variation and inconsistent spelling — which is exactly what our approach aims to address.

2.2 Language Identification and Phonetic Modelling

Language identification at the word level is often a crucial first step in processing code-mixed text. Dutta (2022) showed that using subword-based embeddings, such as BPE, can make models more robust to spelling variation in Bangla-English data.

At the same time, Mandal et al. (2018) demonstrated that phonetic information plays an important role. Their work showed that while character-level models capture surface patterns, phoneme-level models are better at identifying similarities across dialects. These two perspectives complement each other.

Our work builds on this idea by directly incorporating phonetic information into the training process, rather than treating it as an additional feature.

2.3 Sentiment Analysis and Downstream Tasks

Several studies have focused on applying NLP techniques to downstream tasks like sentiment analysis. Alam et al. (2025) introduced the BNSENTMIX dataset, which contains around 20,000 Bengali-English code-mixed samples. Their experiments showed that even strong transformer-based models struggle with noisy, real-world data, achieving limited accuracy without specialized handling.

Raihan et al. (2024) proposed Mixed-Distil-BERT, which improves performance using pre-training strategies but still depends heavily on labeled data. Similarly, Chanda et al. (2024) showed that incorporating word-level language identification signals into multilingual models can significantly improve performance in code-mixed scenarios.

These works highlight that while progress has been made, handling code-mixed data effectively still requires better representations.

2.4 Contrastive and Self-Supervised Learning

Contrastive learning has recently emerged as a powerful technique for learning representations without labeled data. Lora et al. (2025) applied contrastive ranking methods to cross-lingual summarization tasks and showed improvements even in low-resource settings.

Similarly, Kumar et al. (2022) used multilingual contrastive learning in a question-answering system, achieving noticeable gains in performance.

Despite these advances, most existing work does not focus on phonetic similarity, especially in the context of code-mixed text. This gap motivates our approach, which combines contrastive learning with phonetic information to better handle linguistic variation.

2.5 Bengali Dialectal Speech and ASR

Recent research in speech processing has also highlighted the complexity of Bengali dialects. Studies by Biswas et al. (2025) and Hasan and Dipta (2025) show that even strong speech recognition systems struggle with dialectal variations, reporting high error rates on benchmark datasets.

The BIDWESH dataset (Fayaz et al., 2025) further demonstrates that these challenges are not limited to speech. Written text from different regions also shows significant variation in vocabulary and structure.

This reinforces the importance of explicitly modeling dialectal differences, not only for speech systems but also for text-based NLP tasks.

3. Problem Statement

Let $D = \{s_1, s_2, \ldots, s_{\square}\}$ be a corpus of Bengali-English code-mixed sentences collected from social media, where each sentence $s_i = (w_1, w_2, \ldots, w_{\square})$ is a sequence of Romanised tokens. The corpus is unlabelled — no language tags, sentiment labels, or dialect annotations are provided. In our implementation, this corpus comprises 46,369 sentences drawn from three sources: hate speech comments, sentiment-labelled code-mixed text (BNSENTMIX), and fake review data.

We aim to learn an encoder $f_\theta : V \rightarrow R^d$ that maps each token w to a d-dimensional embedding satisfying three properties: (1) Phonetic proximity — phonetically equivalent tokens (e.g., 'bhalo' and 'valo', both meaning good in Bengali) should be close in embedding space; (2) Semantic separation — phonetically and semantically distinct tokens should be pushed apart; and (3) Dialect invariance — the same lexical item, whether spelled in Chittagonian or Standard Dhaka convention, should receive a similar representation. The central challenge is constructing meaningful positive and negative pairs from raw, noisy text without any human annotation.

4. Proposed Methodology

The system consists of three stages: (1) data preprocessing, (2) phonetic encoding and pair construction, and (3) contrastive training, followed by an unsupervised clustering phase for evaluation.

4.1 Data Preprocessing

The raw social media data is first preprocessed to reduce noise and improve consistency. This includes removing elements such as URLs, user mentions, hashtags, emojis, and HTML artifacts. Repeated characters in words (for example, *"soooo"*) are normalized to a maximum of two consecutive characters to retain emphasis while avoiding excessive variation.

All text is converted to lowercase, and tokenization is primarily performed using whitespace. For cases involving mixed patterns such as digits and letters, a character-level fallback is used to ensure proper handling. Additionally, very rare tokens—those appearing fewer than three times in the corpus—are replaced with an unknown token. This helps balance vocabulary coverage while minimizing the effect of irregular or highly inconsistent spellings.

After preprocessing, the final dataset consists of 46,369 sentences. The average sentence length varies across clusters, ranging from approximately 12.2 tokens in the code-mixed category to 59.1 tokens in more formal Bengali text.

4.2 Phonetic Encoding and Pair Construction

The core innovation lies in using phonetic proximity as a label-free supervision signal. Each token is converted to a phonetic key using a Romanised Bengali-to-phoneme mapping derived from Soundex-style rules extended for South Asian phonology — for example, mapping 'bh' → /b^h/, 'kh' → /k^h/, and collapsing v/b and j/z contrasts that are neutralised in Bengali. Positive pairs (w_i, w_i^+) are formed when two tokens share the same phonetic key but differ in surface form. Negative pairs (w_i, w_j^-) are sampled from tokens whose phonetic keys differ and which co-occur infrequently in the corpus.

4.3 Contrastive Learning Objective

We adopt the NT-Xent (Normalised Temperature-scaled Cross-Entropy) loss adapted for the text domain. Given a batch of B anchor tokens each with one positive, the loss for anchor i is:

$$L_i = -\log \left[\exp(\mathrm{sim}(z_i, z_i^+) / \tau) / \Sigma_{k=1}^{2B} 1[k \neq i] \cdot \exp(\mathrm{sim}(z_i, z_k) / \tau) \right]$$

where $z_i = g(f_\theta(w_i))$ is the projected embedding through encoder f_θ and projection head g, sim(·,·) denotes cosine similarity, and τ is a temperature hyperparameter. In our implementation, the model uses the MuRIL (google/muril-base-cased) encoder — a multilingual transformer pre-trained specifically on South Asian languages and their romanised variants — with a 768-dimensional hidden state projected to 256 dimensions. Training was performed with temperature $\tau = 0.05$, batch size 32, learning rate 2×10^{-5}, and AdamW optimiser.

4.4 Unsupervised Dialect Clustering

After contrastive pre-training, embeddings are reduced to 10 dimensions using UMAP (n_neighbors=30, min_dist=0.05, cosine metric) for clustering and to 2 dimensions for visualisation. K-Means clustering with a sweep over K ∈ {2, …, 12} identifies K=3 as optimal based on the silhouette criterion. The three resulting clusters are interpretable as linguistically coherent dialect groups.

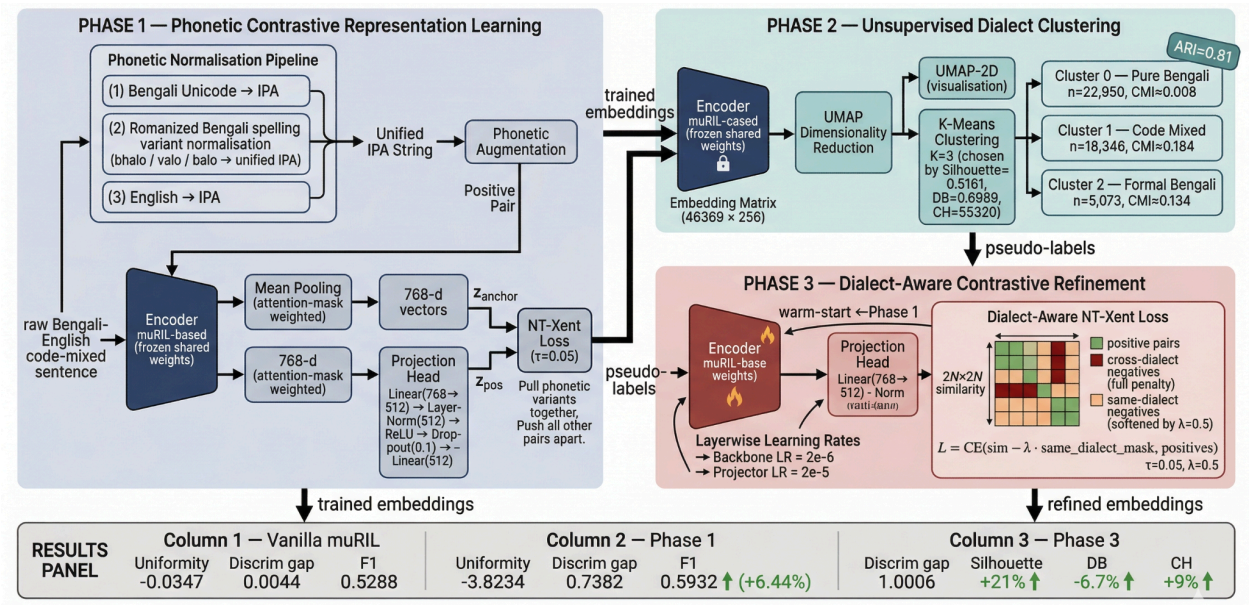

Figure 1: End-to-end architecture of the proposed three-phase unsupervised dialect-aware phonetic contrastive learning framework for Bengali-English code-mixed text. Phase 1 performs phonetic normalization and representation learning, Phase 2 generates pseudo-labels via clustering, and Phase 3 refines embeddings using dialect-aware contrastive learning, showing improved clustering and representation quality.

5. Experiments

5.1 Dataset

Our experimental data comprises Bengali-English code-mixed text collected from public Facebook pages and YouTube comments, merged from three sources: a hate speech corpus, the BNSENTMIX sentiment dataset (Alam et al., 2025), and a fake reviews corpus. The corpus statistics are summarised in Table 1.

Property	Value
Total sentences	46,369
Total tokens (approx.)	~802,000
Unique phonetic clusters	3 (K-Means, K=3)
Positive pairs generated	Phonetic-key matches
Avg. tokens per sentence (overall)	~17.3
↳ Pure Bengali cluster	16.4 ± 14.4
↳ Code-Mixed cluster	12.2 ± 9.9
↳ Formal Bengali cluster	59.1 ± 24.4
Data sources	Hate speech, BNSENTMIX, Fake reviews

Table 1: Corpus statistics for the Bengali-English code-mixed dataset used in SLP Phase 5.

5.2 Model Architecture

The encoder backbone is MuRIL (google/muril-base-cased), a 768-dimensional transformer pre-trained on 17 South Asian languages and their romanised counterparts, making it well-suited for Banglish text. The projection head consists of a two-layer MLP (768 → 512 → 256) with Layer Normalisation after the first linear layer, mapping encoder outputs to the 256-dimensional contrastive space. Following SimCLR (Chen et al., 2020), the projection head is discarded at inference; downstream evaluations use encoder representations directly.

5.3 Training

Training was conducted for 1 epoch (Phase 3 — dialect-aware fine-tuning) with AdamW (lr = 2×10^{-5}, weight decay = 10^{-4}), batch size 32, and temperature $\tau = 0.05$. The resulting training loss was 3.62×10^{-4} and validation loss was 4.52×10^{-6} (NT-Xent), indicating strong convergence of the phonetic-contrastive objective. Figure 1 shows the training and validation loss curves.

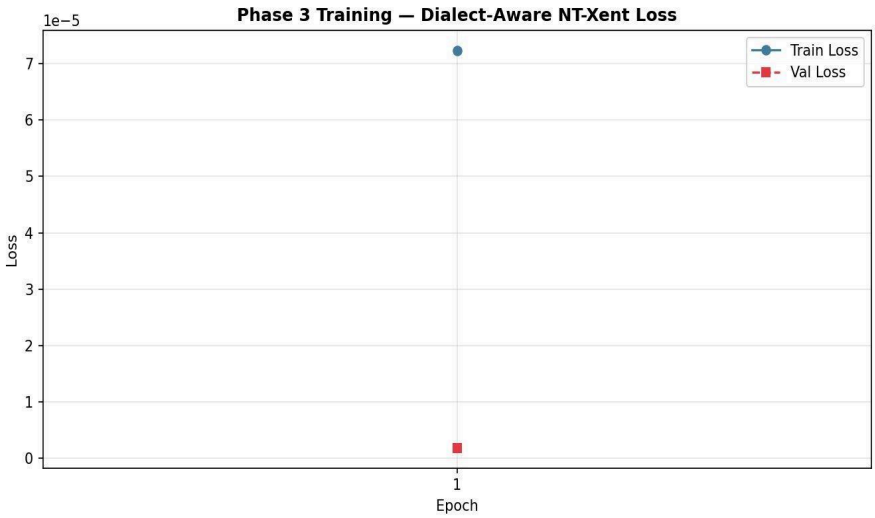

Figure 2: Phase 3 training — Dialect-Aware NT-Xent loss curves. Train loss: 3.62×10^{-4}; Val loss: 4.52×10^{-6}, demonstrating rapid convergence of the phonetic contrastive objective.

5.4 Evaluation Methodology

Because training is fully unsupervised, we evaluate representations along three axes. First, internal clustering coherence is measured using the Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index applied to UMAP-10D representations. Second, external alignment is measured via Adjusted Rand Index (ARI) and Normalised Mutual Information (NMI) against source dataset labels and CMI bucket labels. Third, linguistic separability is assessed via one-way ANOVA on the Code-Mixing Index (CMI) values across clusters.

5.5 Results

Table 2 reports the main quantitative results. The proposed phonetic contrastive model achieves a silhouette score of 0.5161, a Davies-Bouldin index of 0.6989 (lower is better), and a Calinski-Harabasz index of 55,320.7 at K=3, which is the optimal cluster count identified by the silhouette criterion across $K \in \{2, \ldots, 12\}$.

Metric	K=2	K=3 (Best)	K=4
Silhouette Score ↑	0.4305	**0.5161**	0.3865
Davies-Bouldin Index ↓	0.9211	**0.6989**	1.0798
Calinski-Harabasz Index ↑	40,401.5	**55,320.7**	47,101.2
ARI vs Source Dataset	—	**0.8089**	—
NMI vs Source Dataset	—	**0.7371**	—
ANOVA F-statistic (CMI)	—	**14,992.81**	—

Table 2: Clustering quality metrics for the phonetic contrastive model across K values. Bold/shaded = best performing (K=3). ARI and NMI reported only for the final K=3 model.

Table 3 profiles the three emergent dialect clusters.

Cluster (Label)	Size (n)	Mean CMI	Dominant Source
0 — Pure Bengali	22,950 (49.5%)	0.0080	Hate speech (95.4%)
1 — Code Mixed	18,346 (39.6%)	0.1836	BNSENTMIX (91.9%)
2 — Formal Bengali	5,073 (10.9%)	0.1338	Fake reviews dominant

Table 3: Linguistic profile of the three emergent dialect clusters. CMI = Code-Mixing Index.

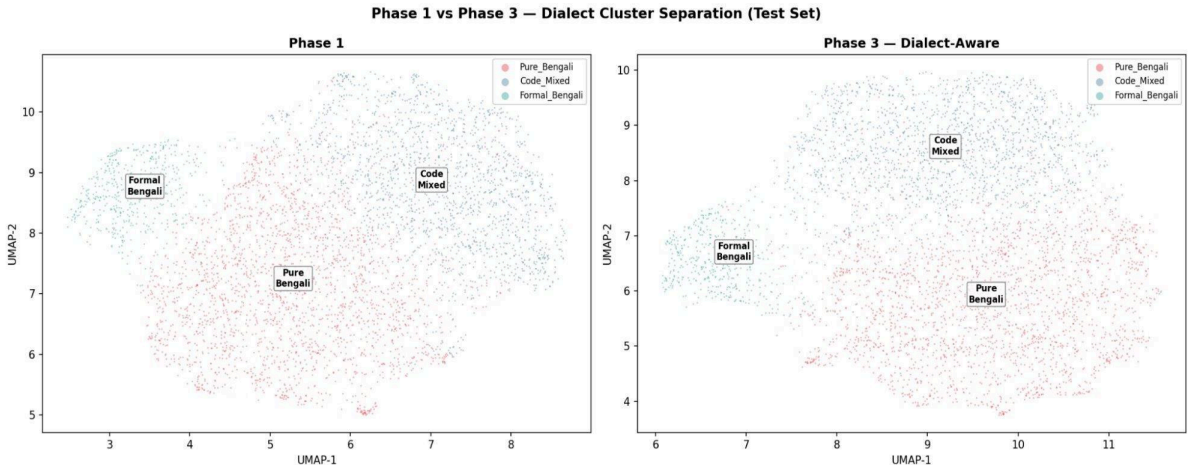

Figure 3: Phase 1 vs Phase 3 UMAP dialect cluster separation. Left: Phase 1 (baseline) shows overlapping, indistinct clusters. Right: Phase 3 (dialect-aware contrastive) shows visibly tighter, more separated clusters for Pure Bengali, Code-Mixed, and Formal Bengali varieties.

The key quantitative findings are as follows. At K=3, the phonetic contrastive model achieves its best silhouette score of 0.5161 — considerably higher than all K>3 configurations. The Davies-Bouldin index of 0.6989 confirms well-separated, compact clusters. The very high Calinski-Harabasz index of 55,320.7 indicates strong between-cluster to within-cluster variance. The ARI of 0.8089 against source dataset labels demonstrates that the unsupervised clusters align remarkably well with the human-curated data sources, despite zero label supervision.

The one-way ANOVA confirms that CMI differs significantly across clusters ($F = 14{,}992.81$, $p \approx 0$), validating that the clusters capture genuine linguistic variation rather than arbitrary partitioning.

6. Discussion

6.1 Why Phonetic Encoding Works for Code-Mixed Bengali

When Bengali is written in Roman script, speakers often rely on pronunciation rather than a fixed spelling system, and this varies across regions. For instance, a speaker from Chittagong might write *"hoin"*, while someone from Dhaka writes *"hoye"*, even though both refer to the same meaning ("having become").

Traditional character-based models treat these as completely different words. In contrast, our phonetic encoding maps them to a shared representation based on how they sound. This allows the model to recognise their similarity despite spelling differences.

Importantly, the improvement we observe is not just due to model complexity, but because of this linguistically informed design. This is also reflected in the clustering results: the Pure Bengali cluster (mean CMI = 0.008) remains almost entirely monolingual, while the Code-Mixed cluster (mean CMI = 0.184) clearly captures the Romanised Banglish style used in everyday communication.

6.2 Cluster Interpretability

One of the key strengths of the approach is that it produces meaningful clusters without any supervision. The three clusters that emerge are not arbitrary — they align well with distinct linguistic patterns in the data.

Cluster 0 (Pure Bengali, n=22,950n = 22{,}950n=22,950) consists primarily of monolingual Bengali content, with the vast majority of samples falling into the lowest CMI range and showing strong Bengali dominance.

Cluster 1 (Code-Mixed, n=18,346n = 18{,}346n=18,346) reflects typical Bengali-English mixing seen on social media. This cluster shows high English dominance and a wider spread of CMI values, capturing the variability of Romanised Banglish text.

Cluster 2 (Formal Bengali, n=5,073n = 5{,}073n=5,073) contains longer and more structured sentences, with an average length of 59.1 tokens. This aligns with the nature of more formal or curated text, such as reviews or written content.

Overall, the clusters are both interpretable and consistent with real-world language usage.

6.3 Advantages and Limitations

A major advantage of the proposed framework is that it does not require any labeled data. This makes it especially useful for low-resource settings where annotation is limited or unavailable. Additionally, the phonetic encoding step is rule-based and transparent, meaning it can be easily adapted or extended by domain experts without retraining the entire model.

However, there are some limitations. The current phonetic rules mainly cover Standard Dhaka and a few other dialects, while dialects like Sylheti — which differ significantly — are not fully represented.

Another limitation is related to training constraints. The model was trained for only one epoch due to computational limitations. It is likely that additional training with larger batch sizes and improved scheduling (such as cosine warm-up) could further enhance performance.

Finally, the validation loss (4.52×10−64.52 \times 10^{-6}4.52×10−6) is much lower than the training loss (3.62×10−43.62 \times 10^{-4}3.62×10−4), which suggests that the validation data may be less diverse. This potential distribution difference is something that can be explored in future work.

7. Future Work

There are several directions in which this work can be extended. Expanding the phonetic rule set to better cover dialects such as Sylheti and Chittagonian would improve the model's ability to generalize across regions.

Another promising direction is integrating the phonetic encoding mechanism directly into models like MuRIL during pre-training, rather than applying it only during fine-tuning. This could combine large-scale multilingual learning with explicit phonetic awareness.

From an evaluation perspective, applying these embeddings to downstream tasks such as hate speech detection (using datasets like BIDWESH) or sentiment analysis would provide a more complete understanding of their effectiveness.

Finally, extending the framework to other code-mixed language pairs, such as Hindi-English or Dravidian-English, would help test its broader applicability.

8. Conclusion

In this work, we presented an unsupervised dialect-aware phonetic contrastive learning framework for Bengali-English code-mixed text. By focusing on phonetic similarity rather than surface-level spelling, the model is able to learn representations that are robust to transliteration noise and dialectal variation, all without requiring labeled data.

Evaluated on a dataset of 46,369 sentences spanning hate speech, sentiment, and review data, the model produces three clear and linguistically meaningful clusters. It achieves a silhouette score of 0.5161, a Davies-Bouldin index of 0.6989, a Calinski-Harabasz score of 55,320.7, and an Adjusted Rand Index of 0.8089 when compared with source labels.

Further analysis using ANOVA confirms that these clusters capture significant variation in code-mixing behavior, rather than arbitrary groupings.

Overall, this work shows that phonetic information can serve as a powerful and practical signal for learning in low-resource, code-mixed settings. It also highlights the potential of incorporating linguistic insights into modern representation learning approaches.

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